

## Autonomous Agentic Multi-Step Reasoning Specification

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Module: WorkerMCPPro

File: backend/app/services/agents/worker\_mcppro\_agent.py

#### 1. Iterative Reasoning Loop

WorkerMCPPro executes an iterative tool-augmented reasoning loop:

Step 1: System prompt initialization with unique question ID and strict output guidelines.

Step 2: LLM invocation with registered tools (retrieve\_context, url\_request).

Step 3: Tool call detection and parallel execution via asyncio.gather.

Step 4: Tool output serialization and insertion into conversation history as 'tool' role messages.

Step 5: Loop continuation until LLM generates final textual answer or max\_iterations (15) is reached.

Step 6: Output post-processing via OutputParserPrompt to enforce schema compliance.

#### 2. Error Recovery and Resilience

If a tool execution raises an exception, the error message is fed back to the LLM within the dialogue history, allowing the model to self-correct, adjust query terms, or select alternative tools.